

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Subject: Classical Mechanics

Course Code: PHYS-3101

Roll No. 48172
Time: 3 Hrs. Mark

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

- Q.1. Answer the following short questions: (6x5=30)
- Use Hamilton's principle of least action to derive Hamilton's equations of motion.
 - Show that the path followed by a particle in sliding from one point to another in the absence of friction in the shortest time is a cycloid.
 - A particle of mass m is sliding on a smooth straight wire which is constrained to rotate in a vertical plane with constant angular speed ω about a fixed point O on the wire. Find the equation of motion of the particle.
 - Show explicitly

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

where $\mathcal{L} = \mathcal{L}(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$.

5. Show that the kinetic energy of two-particle system is

$$\frac{1}{2} M \dot{R}^2 + \frac{1}{2} \mu v^2$$

where $M = m_1 + m_2$, v is the relative speed, and $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ (reduced mass).

6. Show that the total time rate of change of any dynamical variable $A(q_i, p_i, t)$ is given by

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \{A, H\}$$

- Q.2. Answer the following questions. (3x10=30)

- A pendulum of mass m is attached to a massless cart. The cart is attached with a spring of spring constant k and moves along x -axis. Determine the configuration space and find the Lagrange's equations of motion.
- A particle of mass m moves without friction under the action of gravitation on the inner surface of a paraboloid, given by

$$x^2 + y^2 = az$$

Use the method of Lagrange multipliers to determine the Lagrangian and the equations of motion. Show that the particle moves on a horizontal circle in the plane $z = h$, provided that it gets an initial angular velocity. Find this angular velocity.

- Consider Coulomb scattering of charged particles. The Coulomb repulsive potential is $V(r) = \frac{k}{r}$. Start with the equation

$$\theta(r) = \int \frac{l \, dr}{r^2 \sqrt{2\mu \left(E - V - \frac{l^2}{2\mu r^2} \right)}} + \text{constant},$$

and by making suitable substitutions, derive the following expression of the impact parameter

$$b = \frac{k}{2E} \cot \left(\frac{\varphi}{2} \right)$$

where φ is the scattering angle and E is the total energy of the particle.

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This principle states that the time integral of Lagrangian function taken between two configurations of a system has extremum value.

HAMILTON'S PRINCIPLE FOR LEAST ACTION:

we know that $I = \int_{t_1}^{t_2} dt (L)$ i.e. $L = p \dot{q}_i - H$

$$I = \int_{t_1}^{t_2} dt (p_i \dot{q}_i - H)$$

$$\delta I = \int_{t_1}^{t_2} dt \left\{ (\delta p_i) \dot{q}_i + p_i (\delta \dot{q}_i) - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i - \frac{\partial H}{\partial t} \delta t \right\}$$

Last term in the integrand is zero because there is no variation in time. So

$$\delta I = \int_{t_1}^{t_2} dt \left\{ (\delta p_i) \dot{q}_i + p_i (\delta \dot{q}_i) - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right\}$$

Taking $\int_{t_1}^{t_2} p_i \delta(\dot{q}_i) dt$ (integrating by parts)

$$= \left[p_i \delta q_i \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{p}_i \delta q_i dt$$

$$\therefore \delta q(t_2) = 0 \quad \delta q(t_1) = 0$$

$$\text{So } \delta I = \int_{t_1}^{t_2} dt \left\{ (\delta p_i) \dot{q}_i - \dot{p}_i (\delta q_i) - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right\}$$

$$\delta I = \int_{t_1}^{t_2} dt \left\{ \left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) \delta p_i - \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right) \delta q_i \right\}$$

$$\text{So } \dot{q}_i - \frac{\partial H}{\partial p_i} = 0 \quad \dot{p}_i + \frac{\partial H}{\partial q_i} = 0$$

are the Conditions for least action

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \text{and} \quad \dot{p}_i = - \frac{\partial H}{\partial q_i}$$

are the Conditions for least action.

■ **Problem 6:** Show that path followed by a particle in sliding from one point to another in the absence of friction is the shortest time is cycloid. [PU 13, 15, 16]

The Brachistochrone Problem

Solution

A problem concerned with the minimizing the curve that join two points along which a particle fall under the action of gravity from high point to low point in shortest possible time is called Brachistochrone problem.

Greek words *brachistos* means shortest and *chronos* means time.

If particle is falling with speed v through distance $d\ell$ from point 1 to 2 in time t then by definition

$$v = \frac{d\ell}{dt}$$

$$dt = \frac{d\ell}{v}$$

$$\int dt = \int \frac{d\ell}{v}$$

Integrating,

$$t = \int \frac{d\ell}{v}$$

...(1)

If particle is initially at height y then according to conservation of energy, potential energy will convert into kinetic energy. So

$$\Delta T = \Delta V$$

$$\frac{1}{2}mv^2 = mgx$$

$$v = \sqrt{2gx}$$

The length element of the curve is

$$d\ell = \sqrt{1+y'^2} dx$$

Putting value of v and $d\ell$ in eq.(1)

$$t = \int \frac{\sqrt{1+y'^2}}{\sqrt{2gx}} dx$$

$$t = \int_a^b f(y, y'; x) dx$$

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where $f(y, y'; x) = \frac{\sqrt{1+y'^2}}{\sqrt{2gx}}$

For shortest time, t must be minimum. So differentiating w.r.t. y and y'

$$\frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{2gx(1+y'^2)}}$$

Putting in Euler-Lagrange equation

$$0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{2gx(1+y'^2)}} \right) = 0$$

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{2gx(1+y'^2)}} \right) = 0$$

$$d \left(\frac{y'}{\sqrt{2gx(1+y'^2)}} \right) = (0)dx$$

Integrating,

$$\int d \left(\frac{y'}{\sqrt{2gx(1+y'^2)}} \right) = \int (0)dx$$

$$\frac{y'}{\sqrt{2gx(1+y'^2)}} = a$$

where a is constant of integration.

Squaring both sides and solving for y'

$$\frac{y'^2}{2gx(1+y'^2)} = a^2$$

$$\frac{y'^2}{x(1+y'^2)} = 2ga^2$$

Let $k = \frac{1}{4ga^2}$ then $2ga^2 = \frac{1}{2k}$. So

$$\frac{y'^2}{x(1+y'^2)} = \frac{1}{2k}$$

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where k is a new constant. So

$$2ky'^2 = x(1+y'^2)$$

$$2ky'^2 = x + xy'^2$$

$$(2k-x)y'^2 = x$$

$$y'^2 = \frac{x}{2k-x}$$

$$y' = \sqrt{\frac{x}{2k-x}}$$

$$\frac{dy}{dx} = \sqrt{\frac{x}{2k-x}}$$

$$\int dy = \int \sqrt{\frac{x}{2k-x}} dx$$

$$x = k(1 - \cos \theta)$$

...(1)

$$dx = k \sin \theta d\theta$$

$$y = \int \sqrt{\frac{k(1 - \cos \theta)}{(2k - k + k \cos \theta)}} k \sin \theta d\theta$$

$$= \int \sqrt{\frac{k(1 - \cos \theta)}{(k + k \cos \theta)}} k \sin \theta d\theta$$

$$\text{Let } \cos \theta = u$$

Using substitution of variables

So,

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$$\sqrt{2k-x}$$

$$y' = \sqrt{\frac{x}{2k-x}}$$

$$\frac{dy}{dx} = \sqrt{\frac{x}{2k-x}}$$

$$\int dy = \int \sqrt{\frac{x}{2k-x}} dx$$

Using substitution of variables

$$x = k(1 - \cos \theta) \quad \dots(1)$$

$$dx = k \sin \theta d\theta$$

So,

$$y = \int \sqrt{\frac{k(1 - \cos \theta)}{(2k - k + k \cos \theta)}} k \sin \theta d\theta$$

$$= \int \sqrt{\frac{k(1 - \cos \theta)}{(k + k \cos \theta)}} k \sin \theta d\theta$$

$$= \int \sqrt{\frac{k(1 - \cos \theta)}{k(1 + \cos \theta)}} k \sin \theta d\theta$$

$$= \int \sqrt{\frac{(1 - \cos \theta)^2}{(1 + \cos \theta)(1 - \cos \theta)}} (k \sin \theta d\theta)$$

$$= \int \frac{(1 - \cos \theta)}{\sin \theta} (k \sin \theta d\theta)$$

$$y = k \int (1 - \cos \theta) d\theta \quad \dots(2)$$

$$y = k(\theta - \sin \theta) + c$$

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CLASSICAL MECHANICS

To find c using initial condition at initial point $x = 0$ in eq.(1)

$$0 = k(1 - \cos \theta)$$

$$\cos \theta = 1$$

$$\theta = 0$$

Putting $\theta = 0$ at $y = 0$ in eq.(2)

$$0 = k(0 - 0) + c$$

$$0 = c$$

$$y = k(\theta - \sin \theta)$$

$$\left. \begin{aligned} x &= a(\theta - \sin \theta) \\ y &= a(1 - \cos \theta) \end{aligned} \right\}$$

Thus a set of equations is

These are the parametric equations of cycloid passing through origin. For these values of x and y , the time will be shortest.

Between two points is

A bead slides on a smooth wire that is rotating uniformly about a point O in a horizontal plane (the xy -plane). Describe the motion of the bead (Figure 4.5).

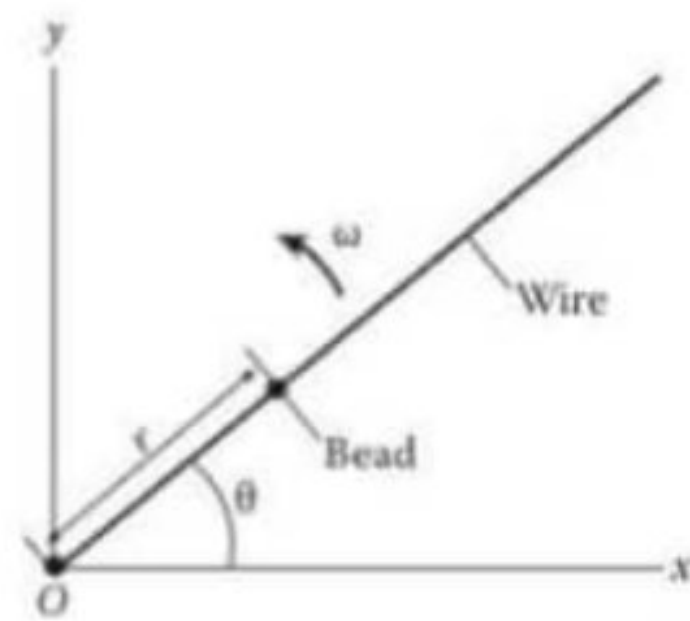


FIGURE 4.5 A bead slides on a smooth wire.

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Solution:

The use of Newton's equation of motion (the second law) would be complicated by the constraint of the wire; we therefore use Lagrange's approach. The kinetic energy is

$$T = (m/2)(\dot{x}^2 + \dot{y}^2) = (m/2)(\dot{r}^2 + r^2\dot{\theta}^2),$$

and the potential energy V is zero. The constraint is that the angular speed is a constant, $\dot{\theta} = \omega$ (a constant), and so the Lagrangian L is

$$L = T - V = T = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2).$$

The Lagrange's equation gives

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = m\ddot{r} - m\omega^2 r = 0,$$

or

$$\ddot{r} = \omega^2 r.$$

This equation can be integrated by multiplying both sides by the integrating factor $\dot{r}$ to give

$$\ddot{r} = \dot{r}\omega^2$$

from which we obtain

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and hence

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$$\frac{d}{dt} \left(\frac{\dot{r}^2}{2} \right) = \omega^2 \frac{d}{dt} \left(\frac{r^2}{2} \right)$$

$$\dot{r}^2 = \omega^2 r^2 + C_1,$$

where C_1 is an integration constant. The last equation can be directly integrated to give

$$r = \sqrt{C_1/\omega^2} \sinh(\omega t + C_2).$$

Prove that $\frac{\partial \dot{\underline{x}}_j}{\partial \dot{q}_i} = \frac{\partial \underline{x}_j}{\partial q_i}$ $\underline{x}_j(q_i, t)$

Proof: Suppose $\underline{x}_j$ is the position vector of j th particle and q_i is the i th generalized coordinates
Then

$$\underline{x}_j = \underline{x}_j(q_1, q_2, \dots, q_n, t)$$

Taking differential, we have

$$d\underline{x}_j = \sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} dq_i + \frac{\partial \underline{x}_j}{\partial t} dt$$

$$\Rightarrow \frac{d\underline{x}_j}{dt} = \sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial \underline{x}_j}{\partial t}$$

$$\Rightarrow \dot{\underline{x}}_j = \sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} \dot{q}_i + \frac{\partial \underline{x}_j}{\partial t}$$

Differentiating w.r.t. $\dot{q}_k$ both sides

$$\frac{\partial \dot{\underline{x}}_j}{\partial \dot{q}_k} = \frac{\partial}{\partial \dot{q}_k} \left(\sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} \dot{q}_i \right) + \frac{\partial}{\partial \dot{q}_k} \left(\frac{\partial \underline{x}_j}{\partial t} \right)$$

$$= \sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} \frac{\partial \dot{q}_i}{\partial \dot{q}_k} + \sum_{i=1}^n \frac{\partial}{\partial q_i} \left(\frac{\partial \underline{x}_j}{\partial \dot{q}_k} \right) \dot{q}_i + 0$$

$\underline{x}_j$ is not
depend upon
 $\dot{q}_k$

$$= \sum_{i=1}^n \frac{\partial \underline{x}_j}{\partial q_i} \delta_{ik} + 0$$

$$= \frac{\partial \underline{x}_j}{\partial q_i} (1) = \frac{\partial \underline{x}_j}{\partial q_i}$$

Since

$$\frac{\partial \underline{x}_j}{\partial \dot{q}_k} = 0$$

$$\delta_{ik} = \frac{\partial \dot{q}_i}{\partial \dot{q}_k}$$

$$\delta_{ik} = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

$T_1 + V_1 = T_2 + V_2$
 $\Rightarrow$ Total energy $T + V$ is conserved.
KINETIC ENERGY OF TWO PARTICLE SYSTEM: Consider two particles m_1 and m_2 at distances x_1 and x_2 from origin. Let m be the centre of mass of the particles whose position vector is R .
 Kinetic energy of the system is given by

$$T = \frac{1}{2} m v^2 + \frac{1}{2} m v_c^2 \quad (1)$$

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Now $R = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$

$M \dot{R} = m_1 \dot{x}_1 + m_2 \dot{x}_2$

$M \dot{R} = m_1 \dot{x}_1 + m_2 \dot{x}_2$

$M \dot{R} = m_1 v_1 + m_2 v_2$

Now relative velocity v is

Given by $v = v_1 - v_2$

From (1) $v_1 = v + v_2$ but in (2)

$M \dot{R} = m_1 (v + v_2) + m_2 v_2$

$M \dot{R} = m_1 v + m_2 v + m_2 v_2$

$M \dot{R} = (m_1 + m_2) v_2 + m_1 v$

$(m_1 + m_2) v_2 = M \dot{R} - m_1 v$

$v_2 = \frac{M \dot{R}}{m_1 + m_2} - \left(\frac{m_1}{m_1 + m_2} \right) v$

Now from (3)

$v_1 = v + v_2$ but in (2)

$M \dot{R} = m_1 (v + v_2) + m_2 v_2$

$M \dot{R} = m_1 v + m_1 v_2 + m_2 v_2$

$M \dot{R} = (m_1 + m_2) v_2 + m_1 v$

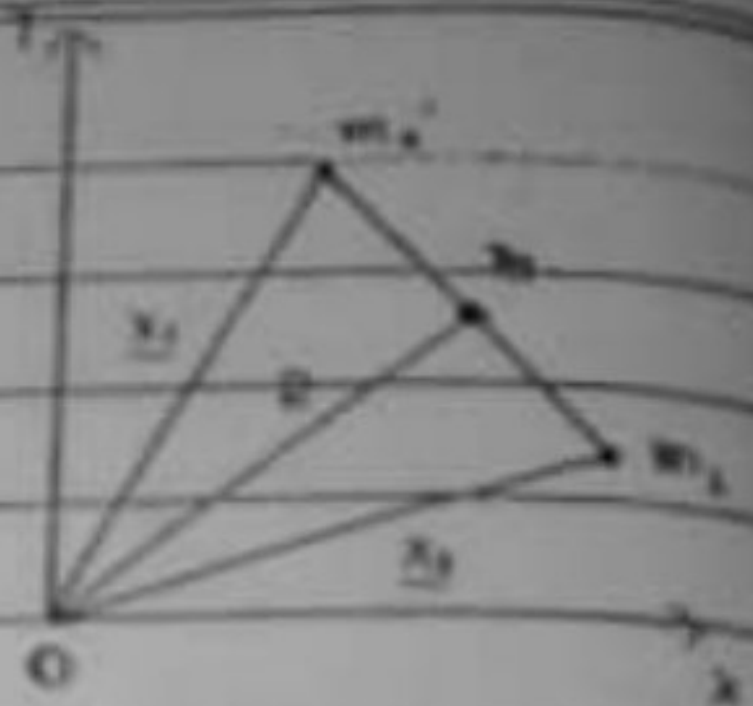
$(m_1 + m_2) v_2 = M \dot{R} - m_1 v$

Now $\underline{\dot{R}} = \frac{m_1 \underline{\dot{x}}_1 + m_2 \underline{\dot{x}}_2}{m_1 + m_2}$

$M \underline{\dot{R}} = m_1 \underline{\dot{x}}_1 + m_2 \underline{\dot{x}}_2$

$M \underline{\dot{R}} = m_1 \underline{\dot{x}}_1 + m_2 \underline{\dot{x}}_2$

$M \underline{\dot{R}} = m_1 \underline{v}_1 + m_2 \underline{v}_2$ (2)



Now relative velocity $\underline{v}$ is

Given by $\underline{v} = \underline{v}_1 - \underline{v}_2$ (3)

From (3) $\underline{v}_1 = \underline{v} + \underline{v}_2$ but in (2)

$M \underline{\dot{R}} = m_1 \underline{v}_1 + m_2 (\underline{v} + \underline{v}_2)$

$M \underline{\dot{R}} = m_1 \underline{v}_1 + m_2 \underline{v} + m_2 \underline{v}_2$

$M \underline{\dot{R}} = (m_1 + m_2) \underline{v}_1 + m_2 \underline{v}$

$(m_1 + m_2) \underline{v}_1 = M \underline{\dot{R}} + m_2 \underline{v}$

$\underline{v}_1 = \underline{\dot{R}} + \left(\frac{m_2}{m_1 + m_2} \right) \underline{v}$ (4)

Again from (3)

$\underline{v}_2 = \underline{v} + \underline{v}_1$ but in (2)

$M \underline{\dot{R}} = m_1 (\underline{v} + \underline{v}_1) + m_2 \underline{v}_2$

$M \underline{\dot{R}} = m_1 \underline{v} + m_1 \underline{v}_1 + m_2 \underline{v}_2$

$M \underline{\dot{R}} = (m_1 + m_2) \underline{v}_2 + m_1 \underline{v}$

$(m_1 + m_2) \underline{v}_2 = M \underline{\dot{R}} - m_1 \underline{v}$

$\underline{v}_2 = \underline{\dot{R}} - \left(\frac{m_1}{m_1 + m_2} \right) \underline{v}$ (5)

but (4) and (5) in (1)

$T = \frac{1}{2} m_1 \left| \underline{\dot{R}} + \left(\frac{m_2}{m_1 + m_2} \right) \underline{v} \right|^2 + \frac{1}{2} m_2 \left| \underline{\dot{R}} - \left(\frac{m_1}{m_1 + m_2} \right) \underline{v} \right|^2$

$= \frac{1}{2} m_1 \left[\underline{\dot{R}}^2 + \left(\frac{m_2}{m_1 + m_2} \right)^2 \underline{v}^2 + 2 \underline{\dot{R}} \left(\frac{m_2}{m_1 + m_2} \right) \underline{v} \right]$

$+ \frac{1}{2} m_2 \left[\underline{\dot{R}}^2 + \left(\frac{m_1}{m_1 + m_2} \right)^2 \underline{v}^2 - 2 \underline{\dot{R}} \left(\frac{m_1}{m_1 + m_2} \right) \underline{v} \right]$

$$v_1 = \frac{R}{m_1+m_2} + \left(\frac{m_2}{m_1+m_2}\right)v \quad (4)$$

Again from (3)

$$v_2 = v + v_1 \quad \text{put in (2)}$$

$$M\dot{R} = m_1(v + v_1) + m_2 v_1$$

$$M\dot{R} = m_1 v + m_1 v_1 + m_2 v_1$$

$$M\dot{R} = (m_1+m_2)v_1 + m_1 v$$

$$(m_1+m_2)v_1 = M\dot{R} - m_1 v$$

$$v_1 = \dot{R} \left(\frac{m_1}{m_1+m_2}\right) \quad (5)$$

put (4) and (5) in (1)

$$T = \frac{1}{2} m_1 \left[\dot{R} + \left(\frac{m_2}{m_1+m_2}\right)v \right]^2 + \frac{1}{2} m_2 \left[\dot{R} + \left(\frac{m_1}{m_1+m_2}\right)v \right]^2$$

$$= \frac{1}{2} m_1 \left[\dot{R}^2 + \left(\frac{m_2}{m_1+m_2}\right)^2 v^2 + 2 \dot{R} \left(\frac{m_2}{m_1+m_2}\right)v \right]$$

$$+ \frac{1}{2} m_2 \left[\dot{R}^2 + \left(\frac{m_1}{m_1+m_2}\right)^2 v^2 + 2 \dot{R} \left(\frac{m_1}{m_1+m_2}\right)v \right]$$

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$$T = \frac{1}{2} \left[m_1 \dot{R}^2 + m_1 \left(\frac{m_2}{m_1+m_2}\right)^2 v^2 + 2 \left(\frac{m_1 m_2}{m_1+m_2}\right) \dot{R} v \right]$$

$$+ m_2 \dot{R}^2 + m_2 \left(\frac{m_1}{m_1+m_2}\right)^2 v^2 + 2 \left(\frac{m_1 m_2}{m_1+m_2}\right) \dot{R} v$$

$$= \frac{1}{2} \left[(m_1+m_2) \dot{R}^2 + \frac{m_1 m_2}{(m_1+m_2)} v^2 \right]$$

$$= \frac{1}{2} \left[(m_1+m_2) \dot{R}^2 + \frac{m_1 m_2}{m_1+m_2} v^2 \right]$$

$$= \frac{1}{2} \left[(m_1+m_2) \dot{R}^2 + \frac{1}{\left(\frac{1}{m_1} + \frac{1}{m_2}\right)} v^2 \right]$$

$$T = \frac{1}{2} (m_1+m_2) \dot{R}^2 + \frac{1}{2} \mu v^2 \quad \left(\mu = \frac{1}{\frac{1}{m_1} + \frac{1}{m_2}} \right)$$

$$\left(T = \frac{1}{2} M \dot{R}^2 + \frac{1}{2} \mu v^2 \right)$$

$T =$ (K.E of centre) + (K.E due to relative motion)
of mass

CONSTRAINTS: the coordinates which restrict the



According to Jacobi's identity

$$0 = [[u, v], H] + [[v, H], u] + [[H, u], v] \quad \text{so} \quad \frac{d[u, v]}{dt} = 0$$

Or $[u, v] = \text{constant}$ of motion or integral of motion. Hence we have $[u, v] = \text{constant}$.

Poisson's Brackets and Hamilton Equations of Motion

Like Hamilton equations, Poisson's brackets are invariant under canonical transformations. As a result of the canonical invariance of Poisson brackets, the relation so obtained will also be invariant in form under a canonical transformation. Suppose a function $u = u(q_j, p_j, t)$, its total time derivative by the use of Hamilton equations of motion is as under:

$$\begin{aligned} \frac{du}{dt} &= \sum_j \frac{\partial u}{\partial q_j} \dot{q}_j + \sum_j \frac{\partial u}{\partial p_j} \dot{p}_j + \frac{\partial u}{\partial t} \\ \frac{du}{dt} &= \sum_j \frac{\partial u}{\partial q_j} \frac{\partial H}{\partial p_j} - \sum_j \frac{\partial u}{\partial p_j} \frac{\partial H}{\partial q_j} + \frac{\partial u}{\partial t} \quad \left(\because \dot{q}_j = \frac{\partial H}{\partial p_j} \text{ and } \dot{p}_j = -\frac{\partial H}{\partial q_j} \right) \\ \text{or } \frac{du}{dt} &= \sum_j \left[\frac{\partial u}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial u}{\partial p_j} \frac{\partial H}{\partial q_j} \right] + \frac{\partial u}{\partial t} \\ \frac{du}{dt} &= [u, H] + \frac{\partial u}{\partial t} \end{aligned} \quad (7.92)$$

If arbitrary function u is replaced by the canonical variables q_j or p_j , Eq.(7.92) gives when q_j and p_j do not contain t explicitly i.e., $\frac{\partial q_j}{\partial t} = 0$ & $\frac{\partial p_j}{\partial t} = 0$.

$$\begin{aligned} \frac{dq_i}{dt} &= [q_i, H] + \frac{\partial q_i}{\partial t} \\ \frac{dq_i}{dt} &= [q_i, H] + 0 = [q_i, H] \end{aligned} \quad (7.93)$$

And,

$$\begin{aligned} \frac{dp_i}{dt} &= [p_i, H] + \frac{\partial p_i}{\partial t} \\ \frac{dp_i}{dt} &= [p_i, H] + 0 = [p_i, H] \end{aligned} \quad (7.94)$$

These two Eqs. (7.93) and (7.94) can also be obtained from definition of Poisson brackets with $V = H$ and $u = q_i$ or p_i . It is easy to note that Eqs. (7.93) and (7.94) are identical with Hamilton equations, so canonical equations of motion are implicit in the properties of Poisson brackets.

Lagrangian is, $L = T - V = ma^2\dot{\theta}^2(1 - \cos\theta) - mga(1 + \cos\theta)$

Lagrange equation for coordinate θ is, $2a(1 - \cos\theta)\ddot{\theta} + a\sin\theta\dot{\theta}^2 - g\sin\theta = 0$

Example 2.30: A pendulum of mass m is attached to a massless cart. The cart is attached with a spring of spring constant k and moves along x -axis. Determine the configuration space and find Lagrange equations of motion.

Solution: Generalized coordinates are x and θ , where x is the displacement of the cart relative to the equilibrium extension a of the spring, and θ is the angle the pendulum arm makes with respect to the vertical. Let the Cartesian coordinates of the pendulum bob be (x_1, y_1) . Therefore,

$$x_1 = x + l\sin\theta, \quad y_1 = -l\cos\theta$$

Kinetic energy is,

$$\begin{aligned} T &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x}_1^2 + \dot{y}_1^2) \\ &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\{(\dot{x} + l\cos\theta\dot{\theta})^2 + (l\sin\theta\dot{\theta})^2\} \\ &\Rightarrow T = \frac{1}{2}(M + m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{x}\dot{\theta}\cos\theta \end{aligned}$$

Potential energy is,

$$V = \frac{1}{2}kx^2 + mgy_1 = \frac{1}{2}kx^2 - mgl\cos\theta$$

Lagrangian is,

$$L = T - V = \frac{1}{2}(M + m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{x}\dot{\theta}\cos\theta - \frac{1}{2}kx^2 + mgl\cos\theta$$

Lagrange equation for coordinate x is,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \Rightarrow (M + m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = -kx$$

Lagrange equation for coordinate θ is,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow ml\dot{x}\cos\theta + ml^2\ddot{\theta} = -mgl\sin\theta$$

Special case: If both x and θ are small, then equations of motion are linearized to

$$(M + m)\ddot{x} + ml\ddot{\theta} + kx = 0$$

$$ml\dot{x} + ml^2\ddot{\theta} + mgl\theta = 0$$

Practice Problem 2.30: Two beads of mass m are connected by a spring (with spring constant k and natural length l).

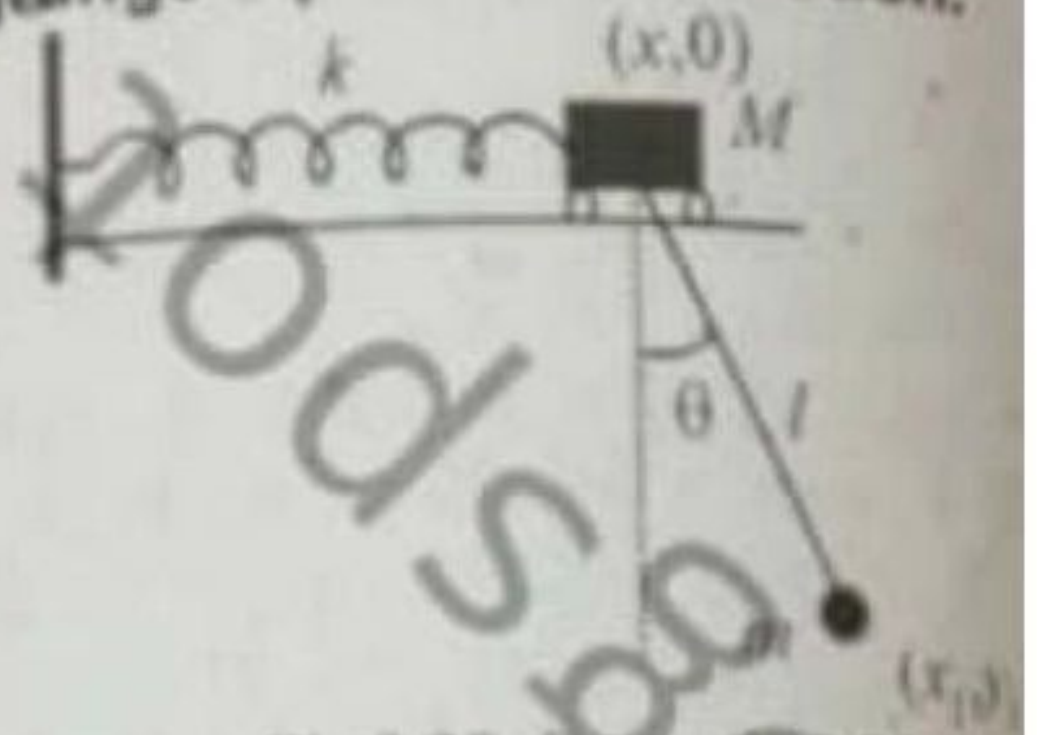


Fig 2.29: Example 2.30

1. Setting up the Lagrangian:

The Lagrangian of a system is the difference between the kinetic energy T and the potential energy V of the system. For a particle moving in a gravitational field on the surface of a paraboloid, we can write the Lagrangian L as:

$$L = T - V$$

where:

- T is the kinetic energy of the particle.
- V is the potential energy of the particle.

For a particle of mass m moving on the surface of the paraboloid $x^2 + y^2 = az$, we'll use cylindrical coordinates (r, θ, z) , where $r = \sqrt{x^2 + y^2}$. The position vector $\mathbf{r}$ of the particle is then $\mathbf{r} = (r, \theta, z)$.

The kinetic energy T is given by $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)$, and the potential energy V is given by $V = mah$, where h is the height of the particle above the base of the paraboloid.

The kinetic energy T is given by $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)$, and the potential energy V is given by $V = mgh$, where h is the height of the particle above the base of the paraboloid.

So, the Lagrangian L becomes:

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2) - mgh$$

2. Finding the Equations of Motion using Lagrange Multipliers:

We need to minimize the action integral $S = \int L dt$ subject to the constraint $f(x, y, z) = x^2 + y^2 - az = 0$. To do this, we introduce Lagrange multipliers λ to the Lagrangian. The equations of motion are then obtained by setting the partial derivatives of the action with respect to the coordinates r , θ , z , and λ equal to zero.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) - \frac{\partial L}{\partial z} = 0$$



$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) - \frac{\partial L}{\partial z} = 0$$

$$f(x, y, z) = 0$$

Solving these equations will give us the equations of motion for the system.

3. Circular Motion:

If the particle is moving in a horizontal circle in a plane $z = h$, it means $\dot{z} = 0$ and $z = h$ remains constant. We can then solve for r and θ in terms of time to find the equation of the circle.

4. Finding Angular Velocity:

The angular velocity ω of the particle can be found by taking the time derivative of θ . Once you have $\theta(t)$, you can find $\omega = \frac{d\theta}{dt}$.

This approach covers the basics of classical mechanics and the method of Lagrange multipliers applied to constrained motion, as well as solving for circular motion and finding angular velocity.

Example 4.19:- Consider the motion of particle in repulsive coulomb potential,

$$V(r) = -\frac{k}{r}$$

Write down the Lagrangian in polar coordinates and integrate the equation of motion to derive

$$\theta(r) = \int \frac{l dr}{r^2 \sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} + \text{constant}$$

where E is total energy and l is angular momentum.

Now change variables $u = \frac{1}{r}$ to derive equation of conic as

$$\frac{p}{r} = 1 + e \cos \theta$$

Solution:- Total energy of two body problem is,

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{l^2}{2\mu r^2} + V(r) \Rightarrow \dot{r}^2 = \frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2} \Rightarrow \dot{r} = \sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}$$

$$\Rightarrow \frac{dr}{dt} = \sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}} \Rightarrow dt = \frac{dr}{\sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}}$$

Since $\dot{\theta} = \frac{l}{\mu r^2}$, $\frac{d\theta}{dt} = \frac{l}{\mu r^2}$ gives,

$$d\theta = \frac{l}{\mu r^2} dt \Rightarrow d\theta = \frac{l}{\mu r^2} \frac{dr}{\sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}} \Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{\mu^2}{l^2} \left\{ \frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2} \right\}}}$$

$$\Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{2\mu E}{l^2} - \frac{2\mu V}{l^2} - \frac{1}{r^2}}}$$

$$\Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2 r} - \frac{1}{r^2}}}$$

Let $r = \frac{1}{u}$, $dr = -\frac{1}{u^2} du$

$$\theta = \int \frac{-\frac{1}{u^2} du}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2} u - u^2}}$$

$$\Rightarrow \theta = \int \frac{-du}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2} u - u^2}}$$

Comparing above equation with

PUACP

$$\int \frac{dx}{\sqrt{a + bx + cx^2}} = \frac{1}{\sqrt{-c}} \cos^{-1} \left(\frac{-b - 2cx}{\sqrt{b^2 - 4ac}} \right)$$

we find that,

$$\frac{2\mu E}{l^2} = a, \quad \frac{2k\mu}{l^2} = b, \quad c = -1, \quad x = u$$

$$\theta = \cos^{-1} \left\{ \frac{-\left(\frac{2k\mu}{l^2}\right) + 2u}{\sqrt{\left(\frac{2k\mu}{l^2}\right)^2 + 4\frac{2\mu E}{l^2}}} \right\} \Rightarrow \cos \theta = \frac{-\left(\frac{2k\mu}{l^2}\right) \left[1 - \frac{l^2 u}{\mu k}\right]}{\left(\frac{2k\mu}{l^2}\right) \sqrt{1 + \frac{2El^2}{\mu k^2}}}$$

$$\Rightarrow \sqrt{1 + \frac{2El^2}{\mu k^2}} \cos \theta = -1 + \frac{l^2 u}{\mu k} \Rightarrow r = \frac{\frac{l^2 u}{\mu k}}{1 + \sqrt{1 + \frac{2El^2}{\mu k^2}} \cos \theta}$$

$$\Rightarrow r = \frac{p}{1 + e \cos \theta} \Rightarrow \frac{p}{r} = 1 + e \cos \theta$$

where $\frac{l^2}{\mu k} = p$ and $e = \sqrt{1 + \frac{2El^2}{\mu k^2}}$